ANALYSIS REPORT

Track 1 Exact-Paper Open-Cell Repair Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the exact-paper open-cell repair campaign prepared in:

- `doc/reports/campaign_plans/2026-04-13-21-20-53_track1_exact_paper_open_cell_repair_campaign_plan_report.md`

  The campaign executed `6` exact-paper validation runs through the dedicated launcher:
- completed runs: `6`
- failed runs: `0`
- execution window: `2026-04-13 22:07:17+02:00` to `2026-04-13 22:09:59+02:00`
- campaign artifact root:

  `output/training_campaigns/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/` This campaign was designed under the new `Track 1` paradigm:
- the primary objective is exact-paper table closure across Tables `3-6`;
- the campaign winner remains a required bookkeeping artifact;
- however, the winner does not define campaign success by itself.

Objective And Outcome

The campaign had three concrete goals

1. refresh the exact-paper baseline under the new cell-closure-first framing;
2. test whether paper-aligned family subsets can close any currently open

   cells in Tables `3-5`;
3. reduce the number of harmonics still marked

   `not_yet_matched_tables_3_6`. Outcome:

- all `6` runs completed successfully;
- all `6` runs exported ONNX artifacts with `0` failed targets;
- no run closed any new numeric paper-target cell in Tables `3-5`;
- the best campaign runs reduced Table `6` open harmonics from `3` to `2`;
- specifically, harmonic `240` moved from

  `not_yet_matched_tables_3_6` to `partially_matched_tables_3_6`;
- no harmonic became fully closed.

  The campaign therefore improved harmonic-family alignment slightly, but it did not close the real numeric gaps that still keep `Track 1` open.

Ranking Policy

This campaign cannot be ranked primarily by winner `MAPE`, because all runs produced the same winning-family aggregate metric. The explicit serialized campaign policy is therefore:

- primary metric: `harmonic_closed_count_desc`
- first tie breaker: `harmonic_open_count_asc`
- second tie breaker: `met_paper_cell_count_desc`
- third tie breaker: `matched_table6_family_target_count_desc`
- fourth tie breaker: `regression_count_asc`
- fifth tie breaker: `enabled_family_count_desc`
- sixth tie breaker: `run_name`

Interpretation:

- this policy exists only to serialize the required campaign-best artifacts;
- the real `Track 1` interpretation remains the cell-closure matrix below.

Campaign Ranking

Ranked Completed Runs

Rank	Config	Family Scope	Harmonics Open	Harmonics Partial	Met Cells	Regressions
1	`exact_open_cell_paper_family_reference`	6 paper-selected families	2	8	14	0
2	`exact_open_cell_phase_gap_bridge`	5 phase-bridge families	2	8	14	0
3	`exact_open_cell_refresh_full_bank`	all 10 families	3	7	14	0
4	`exact_open_cell_tree_control_bank`	8 tree-heavy families	3	7	14	0
5	`exact_open_cell_low_order_repair`	5 low-order families	3	7	11	3
6	`exact_open_cell_high_order_tree_repair`	3 tree families	4	6	11	3

Config	Winner	Mean Component MAPE [%]	Export Mode	Exported	Fail.
`exact_open_cell_paper_family_reference`	RF	18.369	strict	120	0
`exact_open_cell_phase_gap_bridge`	RF	18.369	strict	100	0
`exact_open_cell_refresh_full_bank`	RF	18.369	continue	200	0
`exact_open_cell_tree_control_bank`	RF	18.369	strict	160	0
`exact_open_cell_low_order_repair`	RF	18.369	strict	100	0
`exact_open_cell_high_order_tree_repair`	RF	18.369	strict	60	0

Campaign Best Run

The explicit bookkeeping winner is

- `exact_open_cell_paper_family_reference`

  It was selected because it achieved:
- the minimum harmonic-open count in the campaign: `2`
- the maximum harmonic-partial count in the campaign: `8`
- zero numeric regressions versus the canonical baseline
- the broadest paper-aligned family coverage among the tied top runs

  Important interpretation:
- this run is the campaign bookkeeping winner;
- it is **not** evidence that `Track 1` is solved;
- it is only the cleanest top-ranked run under the paper-closure-first

  ranking policy.

Cell-Closure Impact

Tables `3-5`

There was no numeric closure progress in the campaign.

The still-open numeric cells after the best run remain:

Table	Harmonics Still Open
Table `3` amplitude RMSE	`0` , `1` , `3` , `81` , `240`
Table `4` phase MAE	`1` , `3` , `162` , `240`
Table `5` phase RMSE	`1` , `3` , `39` , `162` , `240`

Table `6`

The campaign did produce one harmonic-status improvement:

- harmonic `240` moved from `not_yet_matched_tables_3_6` to

  `partially_matched_tables_3_6` Best achieved harmonic status after the campaign:
- fully closed harmonics: `0`
- partially matched harmonics: `8`
- still-open harmonics: `2` -> `0` , `1`

Open Numeric Gaps After The Best Run

Cell	Paper Family	Repository Family	Gap To Paper
Table `3` `A_0` RMSE	`SVM`	`HGBM`	`0.000399`
Table `3` `A_1` RMSE	`RF`	`HGBM`	`1.76e-07`
Table `4` `phi_1` MAE	`LGBM`	`HGBM`	`4.64e-05`
Table `5` `phi_1` RMSE	`HGBM`	`GBM`	`1.01e-05`
Table `3` `A_3` RMSE	`HGBM`	`HGBM`	`7.42e-07`
Table `4` `phi_3` MAE	`HGBM`	`GBM`	`0.003757`
Table `5` `phi_3` RMSE	`HGBM`	`HGBM`	`0.005302`
Table `5` `phi_39` RMSE	`HGBM`	`HGBM`	`0.005648`
Table `3` `A_81` RMSE	`RF`	`RF`	`3.22e-06`
Table `4` `phi_162` MAE	`DT`	`ERT`	`0.012507`
Table `5` `phi_162` RMSE	`ERT`	`ERT`	`0.078591`
Table `3` `A_240` RMSE	`ERT`	`RF`	`1.27e-05`
Table `4` `phi_240` MAE	`DT`	`ERT`	`0.039941`
Table `5` `phi_240` RMSE	`ERT`	`ERT`	`0.177287`

The practical reading is blunt

- low-order gaps at `1` and `3` are now numerically tiny on amplitude, but they are still not closed;
- harmonic `0` remains structurally wrong on amplitude because the repository best family is still `HGBM` instead of the paper `SVM` branch;
- the major blockers remain phase-side gaps at `3`, `162`, and `240`;
- `240` improved only on harmonic-family alignment, not on numeric closure.

Interpretation By Run Family

1. The Refresh Run Confirmed The Baseline

`exact_open_cell_refresh_full_bank` reproduced the current exact-paper baseline exactly:

- `5` Table `3` cells met
- `5` Table `4` cells met
- `4` Table `5` cells met
- `0` fully closed harmonics
- `7` partial harmonics
- `3` open harmonics

This means the campaign did not expose hidden regressions in the canonical full-bank branch.

2. The Two Best Runs Improved Only Harmonic Partiality

`exact_open_cell_paper_family_reference` and `exact_open_cell_phase_gap_bridge` both improved Table `6` status by making harmonic `240` partial instead of fully open.

But neither run

- closed a new numeric cell;
- reduced any open numeric gap below paper target;
- produced a first fully closed harmonic.

3. Two Repair Subsets Regressed Existing Good Cells

`exact_open_cell_low_order_repair` and `exact_open_cell_high_order_tree_repair` each regressed `3` previously met cells versus the canonical baseline.

This matters because it shows that family narrowing by itself is not enough. The exact-paper branch is sensitive to subset composition, and some supposedly targeted subsets destroy already-good cells without repairing the intended open ones.

4. Export Stability Is No Longer The Limiting Factor

All `6` runs completed with

- `0` failed ONNX exports
- `0` matched reference relative paths
- no surrogate-export requirement in this campaign

  So the branch is operationally stable. The blocker is no longer exportability. The blocker is now real paper-target closure.

Main Conclusions

The campaign supports five conclusions.

1. The New Track 1 Framing Worked

The campaign was readable in the correct format:

- which cells stayed open;
- which harmonics stayed open or partial;
- which subsets regressed the baseline.

  This is materially better than a winner-only campaign story.

2. The Campaign Did Not Solve Any Numeric Paper Gap

No Table `3-5` cell changed from `above_paper_target` to `met_paper_target`.

So, in the strict sense that matters for `Track 1`, the campaign did not close any table cell.

3. Harmonic `240` Improved Only One Level

The best runs moved `240` from fully open to partial. That is real progress, but still insufficient because all three numeric cells at `240` remain above target.

4. The Main Blockers Are Still Structural

The unresolved branch is now clear

- `A_0` still wants the `SVM`-side paper behavior;
- `1` and `3` are very close numerically but still open;
- `162` and `240` remain the major phase-side blockers.

5. Future Work Must Be More Surgical Than Family Subsets Alone

This campaign shows that family-subset selection alone is too blunt. If the next iteration is meant to close real cells, it likely needs a more target-aware exact-paper workflow than the current family-only runner surface.

Recommended Next Actions

The next technically justified steps are

1. keep `exact_open_cell_paper_family_reference` as the bookkeeping winner for this campaign only;
2. keep the canonical `Track 1` reading anchored to the still-open numeric cells listed above;
3. prioritize a next exact-paper iteration that can act more surgically on: `A_0`, `phi_3`, `phi_162`, and the full `240` triplet;
4. treat `1` and `3` amplitude as near-threshold cells that may benefit from a target-aware refinement instead of another coarse family subset;
5. avoid reusing the low-order-only and high-order-tree-only subsets as-is, because both regressed already-good cells.

Artifact References

- Campaign root:

  `output/training_campaigns/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/`
- Campaign leaderboard:

  `output/training_campaigns/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/campaign_leaderboard.yaml`
- Campaign best run YAML:

  `output/training_campaigns/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/campaign_best_run.yaml`
- Campaign best run note:

  `output/training_campaigns/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/campaign_best_run.md`
- Winning validation summary:

  `output/validation_checks/paper_reimplementation_rcim_exact_model_bank/2026-04-13-22-08-40__exact_open_cell_paper_family_reference_campaign_run/validation_summary.yaml`
- Winning model bundle:

  `output/validation_checks/paper_reimplementation_rcim_exact_model_bank/2026-04-13-22-08-40__exact_open_cell_paper_family_reference_campaign_run/paper_family_model_bank.pkl`
- Per-run launcher logs:

  `output/training_campaigns/track1_exact_paper_open_cell_repair_campaign_2026_04_13_21_20_53/logs/`